

padilla-et-al-2015-the-influence-of-different-graphical-disp

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

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Paper Synthesis

Not available for this paper.

Abstract

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Understanding how people interpret and use visually presented uncertainty data is an important yet seldom studied aspect of data visualization applications. Current approaches in visualization often display uncertainty as an additional data attribute without a well-defined context. Our goal was to test whether different graphical displays (glyphs) would influence a decision about which of 2 weather forecasts was a more accurate predictor of an uncertain temperature forecast value. We used a statistical inference task based on fictional univariate normal distributions, each characterized by a mean and standard deviation. Participants viewed 1 of 5 different glyph types representing 2 weather forecast distributions. Three of these used variations in spatial encoding to communicate the distributions and the other 2 used nonspatial encoding (brightness or color). Four distribution pairs were created with different relative standard deviations (uncertainty of the forecasts). We found that there was a difference in how decisions were made with spatial versus nonspatial glyphs, but no difference among the spatial glyphs themselves. Furthermore, the effect of different glyph types changed as a function of the variability of the distributions. The results are discussed in the context of how visualizations might improve decision making under uncertainty.

Introduction (Paper-Level)

Not available for this paper.

General Discussion

Not available for this paper.

Critique

Not available for this paper.